

Complete radio link budget from all earth stations:

Frequency	437.5	MHz		
Orbital Altitude	500	Km		
Elevation Angle	35	Deg		
Distance	815.089	Km		
	-10	dBW		
Spacecraft Antenna Gain	2.2	dBi		
Spacecraft Line Loss	1	dB		
EIRP	-8.8			
	GFSK (2K00 F1D)	CW Morse (500H A1A)	RTTY FSK (515H F1B)	
Free Space Path Loss::	143.4737	143.4737	143.4737	dB
Antenna Pointing Loss	1	1	1	dB
Antenna Polarization Loss:	1	1	1	dB
Atmospheric Loss:	2	2	2	dB
Ionospheric Loss:	1	1	1	dB
Rain Loss:	0	0	0	dB
Isotropic Signal Level at GS	-157.274	-157.274	-157.274	dB
GS Antenna Pointing Loss	0.2	0.2	0.2	dB
GS Antenna Gain	14	14	14	dB
GS Transmission Loss	2	2	2	dB
GS Effective Noise Temperature	635	635	635	K
GS Figure of merit (G/T)	-16.2277	-16.2277	-16.2277	dB/K

Signal Power at Input of LNA	-145.474	-145.474	-145.474	dBW
Boltzman's constant	-228.6	-228.6	-228.6	dbW/K/Hz
Receiver Bandwidth	2000	500	515	Hz
Receiver Noise Power	-167.562	-173.583	-173.454	dBW
SNR at Receiver	22.0883	28.1089	27.98053	dB
Requested SNR	12.8	10	10	dB
Link Margin	9.288301	18.1089	17.9805	dB